

Minhyuk Nam

minhyukn@andrew.cmu.edu ◦ sites.google.com/view/mhnam ◦ (412) 209-5368

Education

Carnegie Mellon University

2022 – present

Ph.D. Candidate in Economics

Expected Completion Date: May 2027

Dissertation: “Dynamic Market Design in Environmental and Energy Economics”

Committee: Robert A. Miller, Ali Shourideh, Akshaya Jha, Nicholas Z. Muller, Karen Clay

Carnegie Mellon University

2024

M.S. in Economics

Sogang University (Seoul, South Korea)

2022

B.A. in Economics; B.Eng. in Computer Science; B.S. in Mathematics, *Magna Cum Laude*

Exchange Program: Lancaster University, 2017

Mandatory Military Active Duty: Republic of Korea Air Force, 2018–2020

Research Interests

Environmental Economics, Industrial Organization, Mechanism Design

Working Papers

- (i) **Free Allocation, Trading Frictions, and Market Efficiency in Emission Markets** *2026*
with Robert A. Miller

The EU ETS initially gave 90% of carbon permits free to incumbents but is now transitioning to full auctioning. We study whether free allocation reduces efficiency by discouraging trading and creating thin markets. Using EU ETS transaction data, we show that firms receiving free permits trade less frequently, concentrating activity among financial intermediaries who account for over half of transactions despite no compliance obligations themselves. We develop a dynamic structural model of production, banking, and trading decisions under participation costs to estimate efficiency losses from misallocation across heterogeneous emitters. Simulating full auctioning counterfactuals, we quantify whether eliminating free allocation improves welfare by expanding liquidity or whether transaction frictions prevent efficiency gains.

- (ii) **Optimal Climate Treaties as Delegation with Externalities: the Case of Dynamic Carbon Production** *2026*
with Ali Shourideh

International climate agreements must delegate emission decisions to countries with private preferences while managing global environmental costs that accumulate over time. We characterize optimal dynamic climate agreements using mechanism design theory with a continuum of heterogeneous countries and stock externalities. We establish two main results. First, optimal agreements exhibit a cutoff structure where high-emission countries bunch at period-specific caps while low-emission countries receive flexibility. Second, we derive necessary and sufficient conditions for when simple “total carbon budget” mechanisms achieve optimality. Our analysis identifies when international climate frameworks should impose annual emission limits versus flexible carbon budgets.

(iii) **Group Composition and Group Decision-Making: Evidence from Municipal Council Meetings in South Korea** 2026

with Jay Euijung Lee and Martina Zanella

[INSERT ABSTRACT — Roughly 80–150 words. Describe the question, approach, and main finding.]

Work in Progress

1. “Compensating Flexibility: How Capacity Markets Distort Battery Storage Dispatch.”

Publications

1. **The Effects of Indoor Temperature and Humidity on Local Transmission of COVID-19 and How It Relates to Global Trends**
with Han June Park, Sung-Gwang Lee, Jeong Suk Oh, Steven Barrett, Soohyung Lee, and Wontae Hwang
PLoS ONE, 2022.
2. **COVID-19 and Job Market: Trends and Comparison with the 2008 Financial Crisis in South Korea**
with Soohyung Lee
Seoul Journal of Economics, 2021.
3. **Does Ramadan Harm Infant Health? Evidence from Ethiopia**
with Soohyung Lee, Daeun Jeong, and Wonmoon Lee
International Economic Journal, 2020.
4. **Impact of the Clean Air Act on Air Pollution and Infant Health: Evidence from South Korea**
with Soohyung Lee and Heesun Yoo
Economics Letters, 2018.

Research Experience

London School of Economics, Research Assistant 2020–2021

Project: Gender Ratios and Group Interaction: Evidence from Local Parliamentary Meetings in South Korea

Seoul National University, Research Assistant for Professor Soohyung Lee 2016–2022

World Bank, Local Short Term Consultant (Data Analyst) 2016

Project: Bilateral Arrangement of Temporary Labor Migration – Lessons from Korea

Teaching Experience

Carnegie Mellon University, Teaching Assistant 2024–present

Ph.D.: Econometrics II, Econometrics III, Advanced Econometrics

MBA: Managerial Economics, Statistical Decision Making

Undergraduate: Principles of Macroeconomics, Environmental Economics,
Regression Analysis, Probability and Statistics for Business Applications

Sogang University, Teaching Assistant 2018–2022

Undergraduate: Urban Economics, Python Programming

Professional Activities

Presentations and Seminars (including scheduled)

2025 DSE Summer School (London, UK)
2024 Pennsylvania Economic Theory Conference (PennState)
2018 Korean Association of Public Finance Meeting (Seoul)

Referee Service

[Add as applicable]

Honors, Fellowships, and Awards

William Larimer Mellon Fellowship , Carnegie Mellon University	2022–present
Dean’s Research Funds , Carnegie Mellon University	2023
Winner , Kaggle Data Analytic Competition (COVID-19 & Digital Learning)	2021
Merit Based Scholarship , Sogang University	2015

Software Skills

Stata, Python, C, C++, MATLAB, SAS, EViews, $\text{\LaTeX}$

Languages

Korean (native), English (fluent)